

OSBORNE V. LOPES

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Summary

Data Engineer with 3+ years of experience building ETL/ELT pipelines and data platforms using Python, SQL, Snowflake, Azure ADF and Kafka, along with production RAG-based tools using LangChain, FastAPI and Azure AI Search. Skilled at designing scalable data models, orchestrating pipeline workflows with Airflow and DBT, and translating raw data into reliable, presentation-ready analytics across finance and enterprise domains.

Experience

ServiceNow

Feb 2026 - Present

Data Engineer

Remote

- Engineered a PySpark ETL pipeline on Delta Lake that ingests 200GB+ of telemetry data daily, reducing data processing time from 4.5 hours to 38 minutes through dynamic partitioning and caching strategies
- Deployed scalable data pipelines through CI/CD workflows using GitHub Actions to ingest and transform incident, case and workflow data across multiple ServiceNow modules, reducing data processing time by 45% and enabling real time operational dashboards
- Built a FastAPI-based microservice with real-time vector search integration using Azure AI Search, enabling sub-200ms similarity searches across 50M+ embedded vectors for document recommendation
- Deployed a fine-tuned LLM chatbot on Azure Cloud Services handling 8,500+ daily employee queries, achieving 94% intent accuracy and reducing IT support ticket volume by 37% within 3 months

Atlas SP

Jan 2025 - Aug 2025

Data Engineer Co-op

New York, NY

- Built ELT ingestion pipelines, both ADF batch-based and Kafka streaming-based, from Azure Blob Storage to Snowflake, transforming data across raw to presentation layers and applying business logic through Alteryx workflows to power downstream Tableau analytics
- Engineered a production RAG-based chatbot using Azure AI Search vector indexes, LangChain and FastAPI, enabling natural language querying of live financial data for internal teams and reducing repetitive analyst requests by 30%
- Designed normalized data models in Snowflake using complex joins and window functions, improving query performance by 20% and enabling cleaner portfolio analytics for executive reporting
- Collaborated with team members through code reviews and PR merges to maintain pipeline quality, while partnering with business stakeholders to define KPIs and requirements for downstream Tableau dashboarding

Zyphron Industries

Jul 2021 - Aug 2023

Analytics Engineer

Mumbai, India

- Designed star-schema semantic models in Power BI with advanced DAX measures to track inventory, fulfillment, and supplier KPIs, developing executive dashboards with red/yellow/green KPI indicators to highlight performance against targets and provide actionable insights to logistics leadership across 5+ business teams
- Ingested API data into the warehouse and built analytics-ready DBT models with DBT tests validating data quality before promoting to presentation-layer dimensional models, orchestrated via Airflow, reducing data quality issues reaching downstream reports by 25%
- Analyzed customer order and delivery performance using incremental DBT models and materialized views, conducting A/B tests on reporting formats to identify SLA breach trends and reduce repeat complaints by 15%
- Reduced inventory shortfalls by 20% by using SQL and PySpark to transform, aggregate, and analyze historical inventory and order data, engineering peak-season demand metrics to improve stockout prediction and inventory planning

Skills

Languages: Python, Java, R, SQL, Pro*C

Databases and Warehousing: Snowflake, Teradata, MongoDB, DynamoDB, Redshift

Cloud Platforms: Azure (ADF, Blob Storage, Synapse, Fabric), AWS (S3, Glue, EMR, Lambda, Athena, Kinesis, IAM), GCP (BigQuery, DataFlow, GCS), Docker

Tools: Airflow, DBT, Apache Spark, Kafka, Databricks, Alteryx, Control-M, Git

Analytical Toolkit: Tableau, PowerBI, Looker, Google Data Studio, QuickSight, Excel

AI and Automation: RAG, Agentic AI, LangChain, HuggingFace, FastAPI, PineCone, LLM, n8n, Prompt Engineering

Education

Northeastern University

Dec 2025

Master of Science, Information Systems

USA

- Coursework: Big Data and Intelligent Analytics, Data Architecture and Business Intelligence, Data Science Methods and Tools

University of Mumbai

Jun 2021

Bachelor of Technology, Information Technology

India

- Coursework: Data Mining and Business Intelligence, Database Management, Cloud Computing Services

Projects

NewsScope | [GitHub](#)

Dec 2024

- Built NewsScope, an industry-specific news aggregator that scrapes, processes and surfaces real-time business news using Python (BeautifulSoup, Scrapy) and DBT on BigQuery
- Designed an end-to-end ETL pipeline orchestrated via Airflow, landing raw data in GCS and transforming in BigQuery, with schema validation using Pydantic 2 and data quality checks via Great Expectations